

Performance Testing

Date	03 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Flask Test Client, Browser Testing
Type of Testing	Functional Testing, Performance Testing
Target Module / API	Crop Prediction Module
Test Environment	Windows 11, Python 3.11, Flask
Test Date	03 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Crop prediction using valid inputs	1	10	Correct crop is displayed
2	Invalid input values	1	10	Validation error shown
3	Multiple prediction requests	10	30	System responds correctly
4	Continuous prediction requests	20	60	Application remains stable

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 sec	1.2 sec	Pass	Fast response
2	Response Time (Max)	< 5 sec	2.3 sec	Pass	Within limit
3	Throughput	High	25 req/sec	Pass	Stable
4	Error Rate	<1%	0%	Pass	No errors
5	CPU Utilization	<80%	42%	Pass	Normal
6	Memory Utilization	<80%	55%	Pass	Stable

Step 4: Observations & Analysis

Key Findings

- Crop prediction is generated quickly.
- The application remained stable during testing.

Bottlenecks Identified

- No major bottlenecks were observed.
- Response time increased slightly under multiple requests.

Optimization Steps Taken

- Optimized the trained ML model loading.
- Validated user inputs before prediction.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

```
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in production.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://192.168.29.234:5000
```